

$$x_i = ih, \; y_j = jh, \; h = 1/N,$$

$$\frac{4u_{ij} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}}{h^2} = f_{ij},$$

$$1\leq i,j\leq N-1,$$

$$A\mathbf{u}=\mathbf{f},\quad A=A^T,\quad \mathbf{z}^TA\mathbf{z}>0\;(\mathbf{z}\neq 0),$$

$$\mathbf{r}=\mathbf{f}-A\mathbf{u}_h,$$

$$\|u_h-u\|_\infty=O(h^2) \quad \text{for this smooth solution.}$$